8 **(a)** $\mathbf{a} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$

Work out.

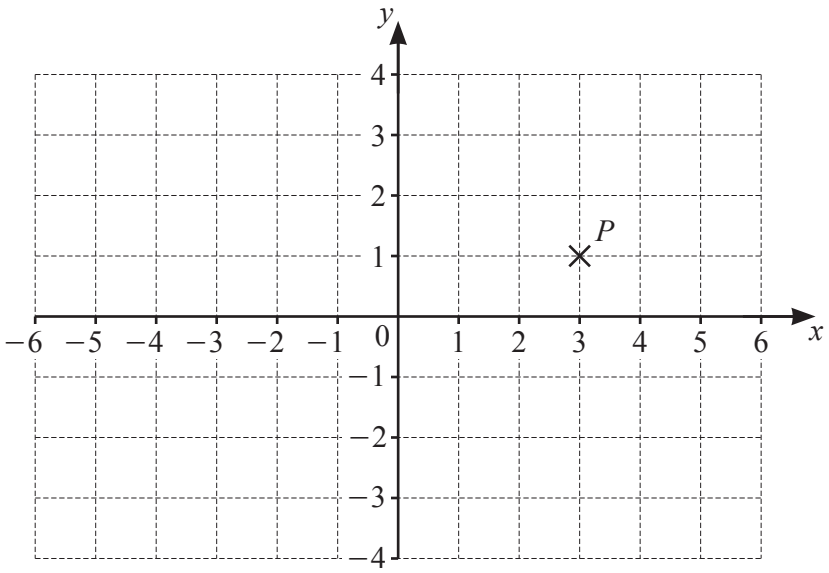
(i) $4\mathbf{a}$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(ii) $2\mathbf{a} - \mathbf{b}$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [2]

(b)



(i) Write down the coordinates of point P.

(..... , ...) [1]

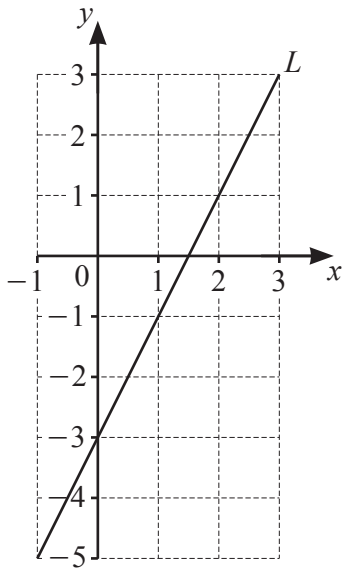
(ii) On the grid, plot point Q at $(-4, 2)$. [1]

(iii) $\overrightarrow{PR} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

On the grid, plot point R. [1]

(iv) On the grid, draw the line $y = 3$. [1]

(c)



Line L is shown on the grid.

(i) Find the equation of line L in the form $y = mx + c$.

$y =$ ..... [2]

(ii) Write down the equation of a line parallel to line L.

$y =$ ..... [1]

Question 9 is printed on the next page.